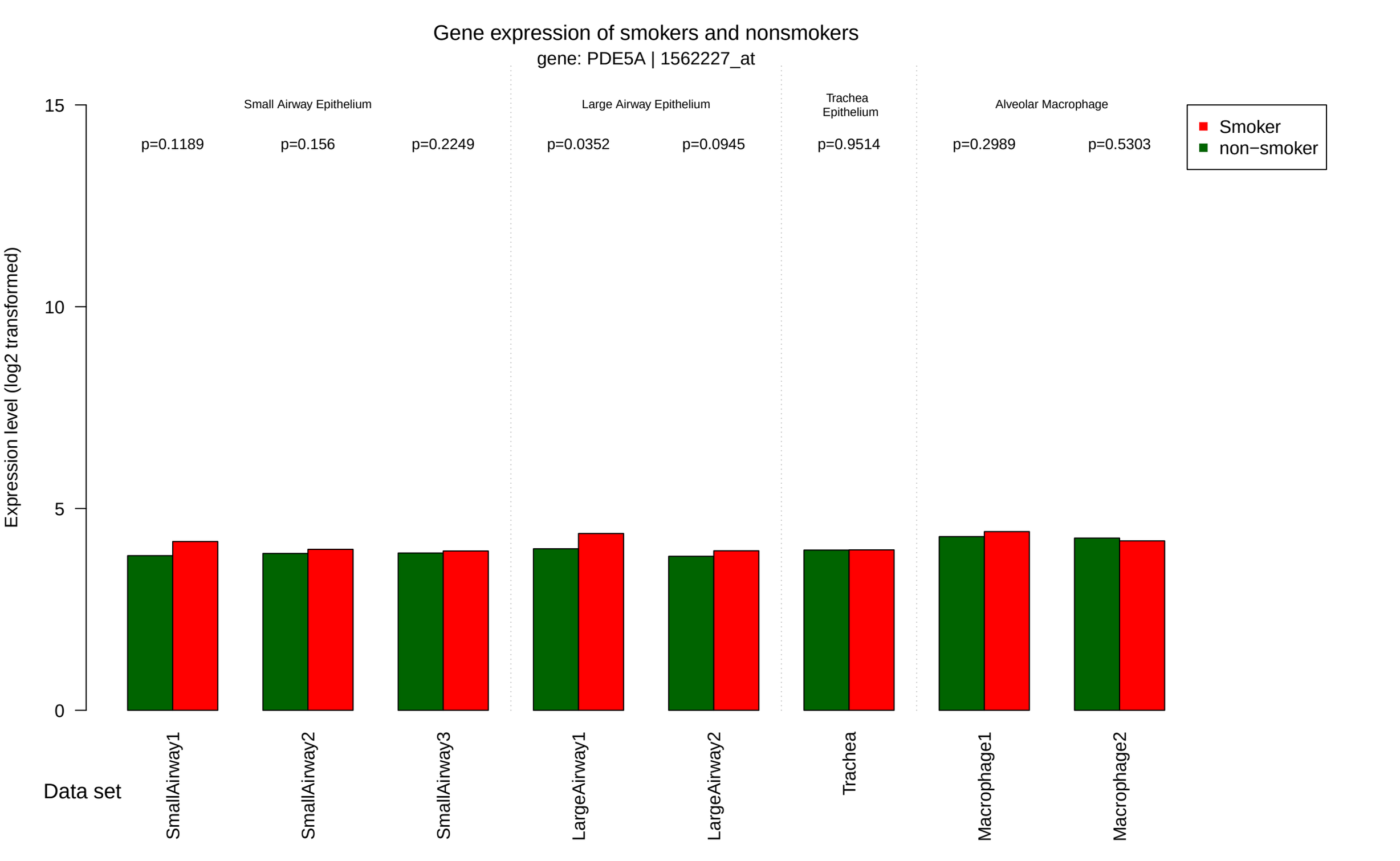
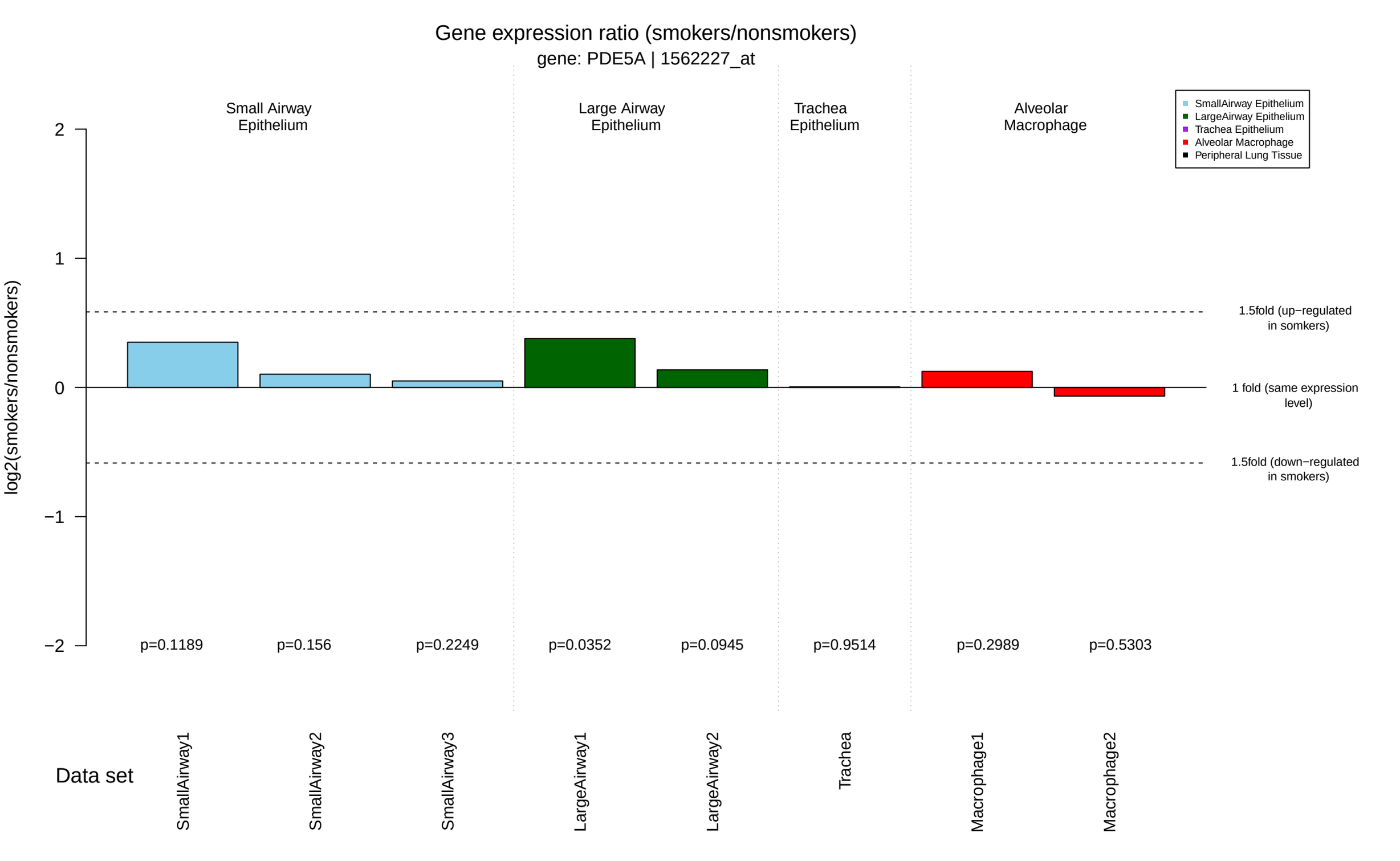
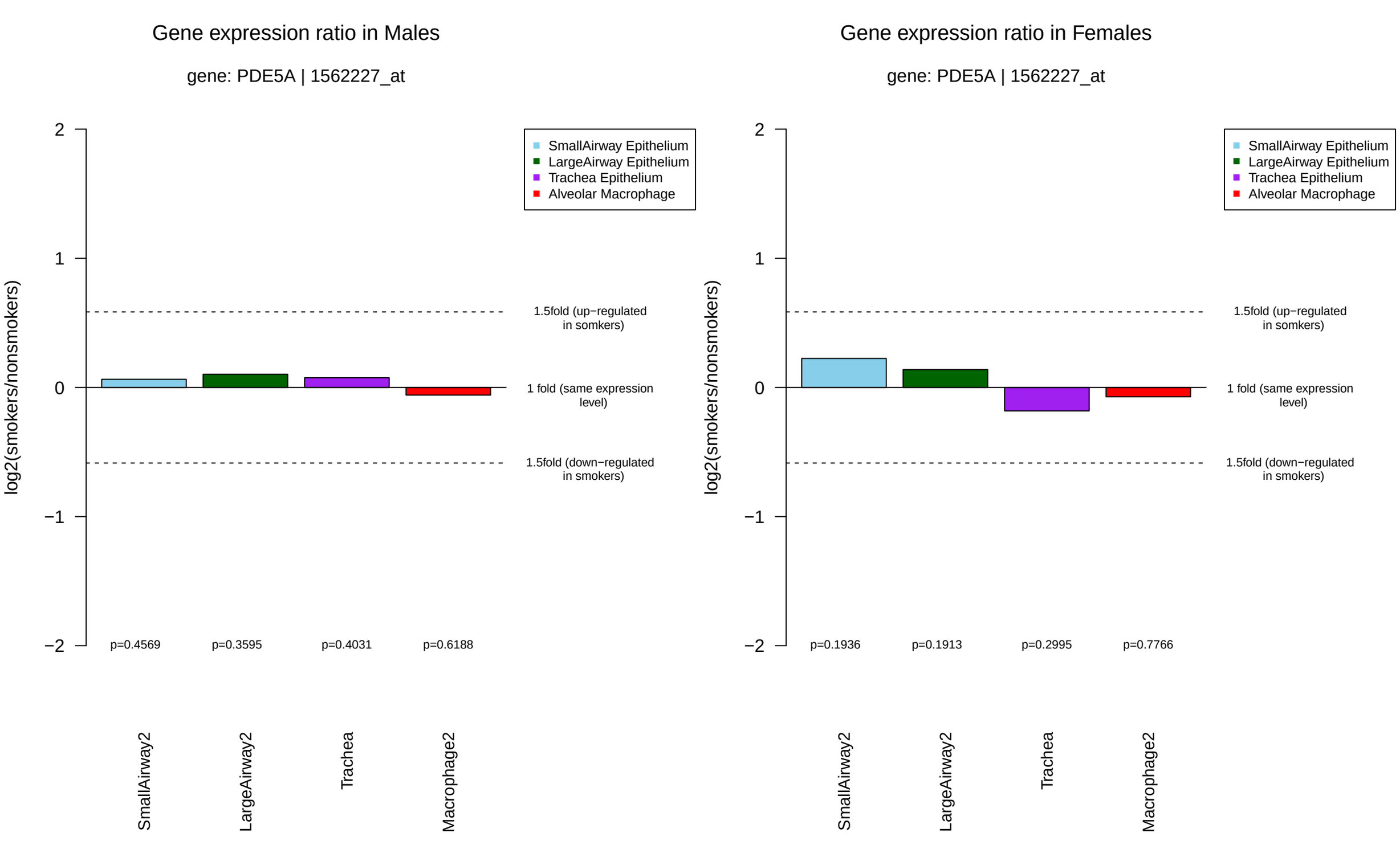
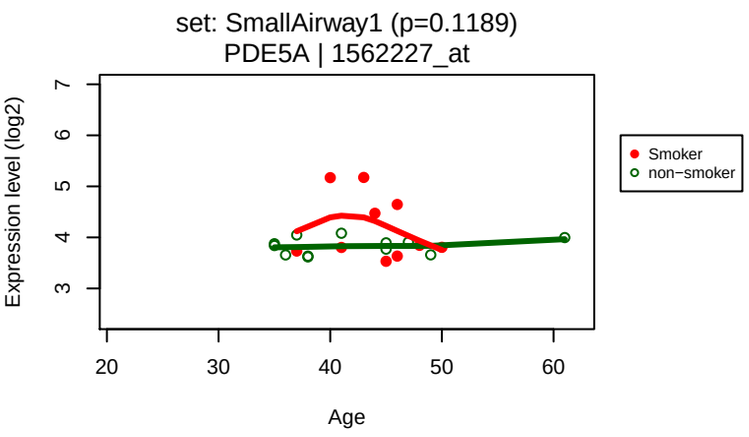
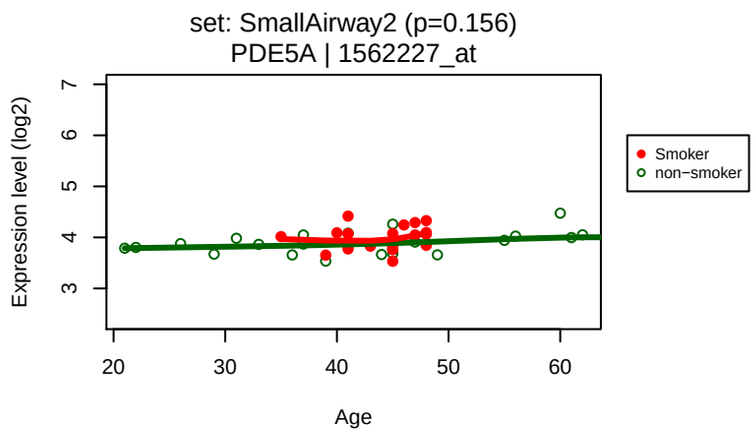
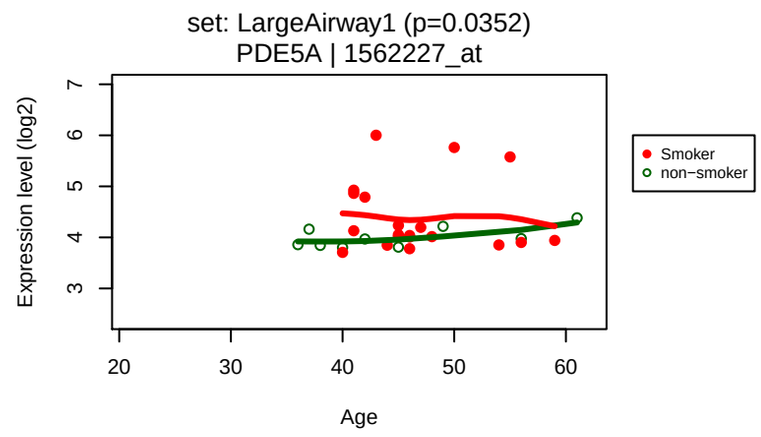

Gene expression of smokers and nonsmokers
gene: PDE5A | 1562227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1189
p=0.156
p=0.2249
p=0.0352
p=0.0945
p=0.9514
p=0.2989
p=0.5303
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

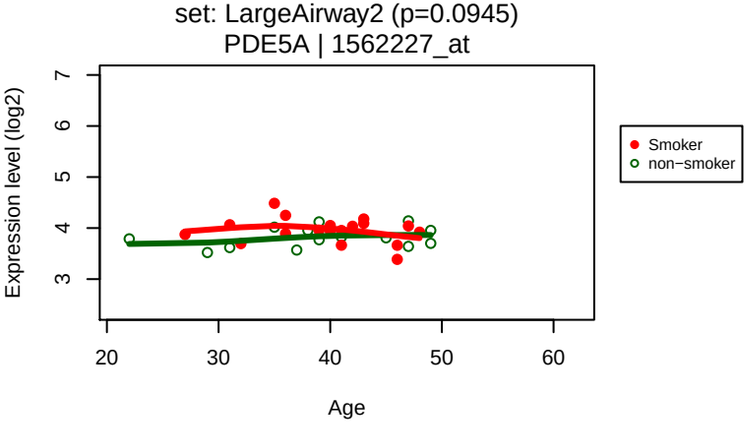

Gene expression ratio (smokers/nonsmokers)
gene: PDE5A | 1562227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1189
p=0.156
p=0.2249
p=0.0352
p=0.0945
p=0.9514
p=0.2989
p=0.5303
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

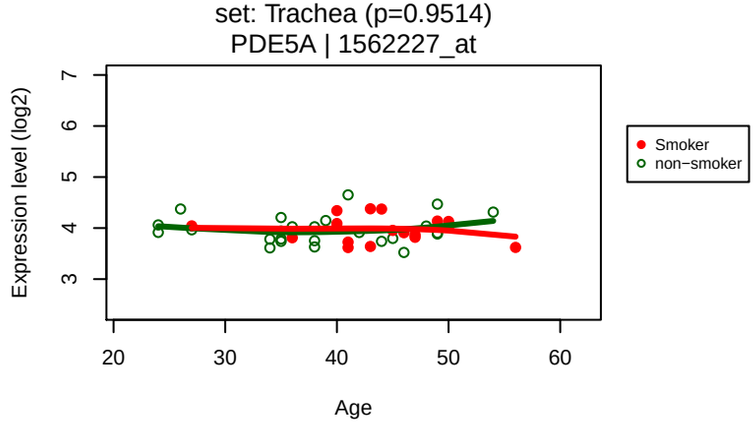

Gene expression ratio in Males
gene: PDE5A | 1562227_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4569
p=0.3595
p=0.4031
p=0.6188
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE5A | 1562227_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1936
p=0.1913
p=0.2995
p=0.7766
SmallAirway2
LargeAirway2
Trachea
Macrophage2

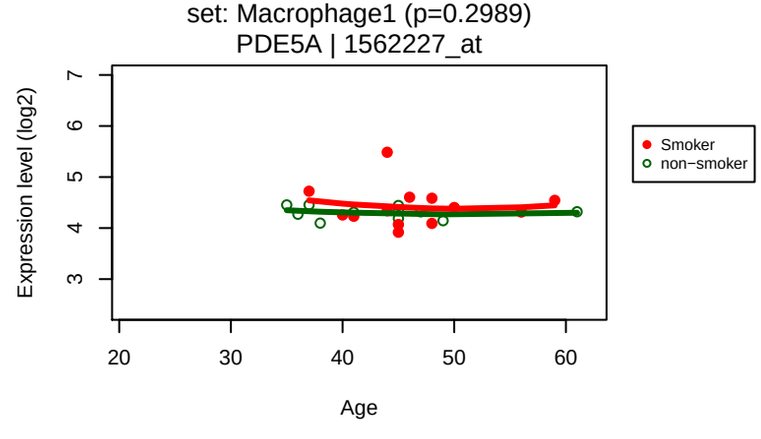

set: SmallAirway1 (p=0.1189)
PDE5A | 1562227_at
Expression level (log2)
Age
Smoker
non−smoker

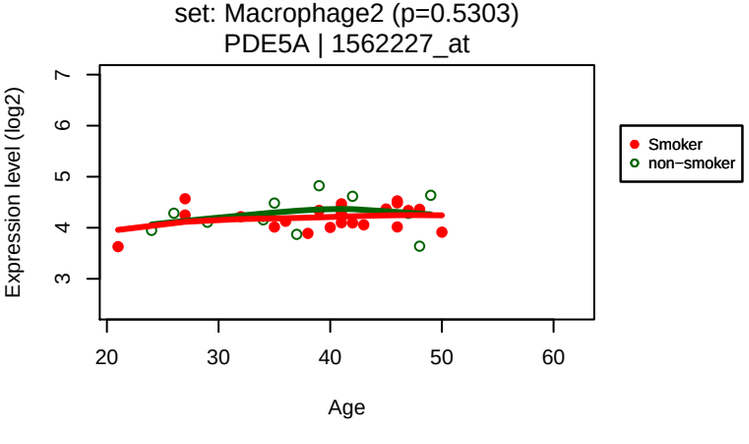

set: SmallAirway2 (p=0.156)
PDE5A | 1562227_at
Expression level (log2)
Age
Smoker
non−smoker

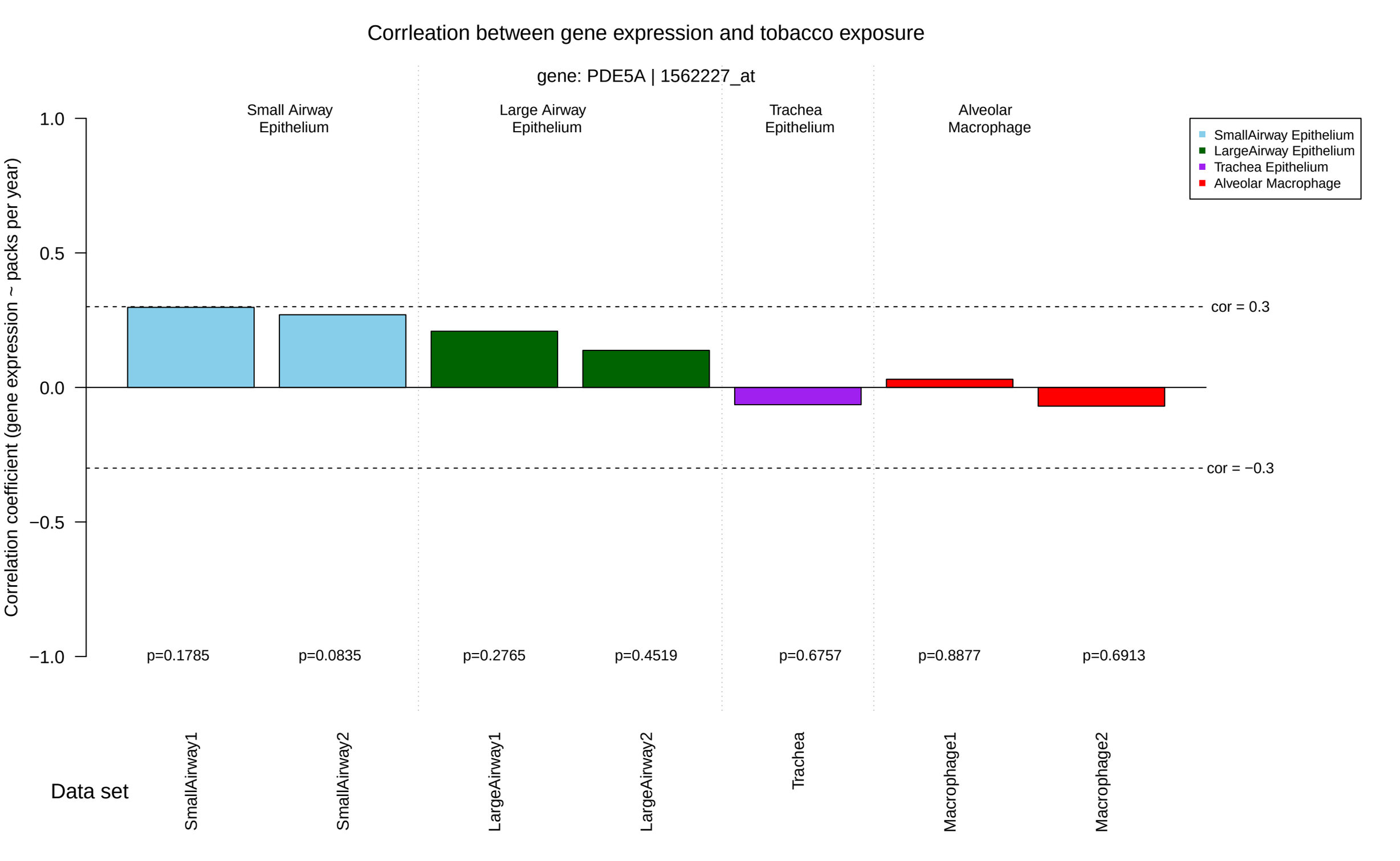

set: LargeAirway1 (p=0.0352)
PDE5A | 1562227_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0945)
PDE5A | 1562227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9514)
PDE5A | 1562227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2989)
PDE5A | 1562227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5303)
PDE5A | 1562227_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE5A | 1562227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1785
p=0.0835
p=0.2765
p=0.4519
p=0.6757
p=0.8877
p=0.6913
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2